

Program overview

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Year 2022/2023
Organization Applied Sciences
Education Bachelor Life Science and Technology

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor LST 2022			
1e jaar BSc LST 2022	1st year BSc LST 2022		
LB1045	Organic Chemistry 1	6	
LB1053	Cell biology & Biochemistry: an introduction	5	
LB1231	Life Sciences	8	
LB1251	Quantitative Molecular Biology	4	
LB1260	Introduction to Algorithmics and Programming using Python	3	
LB1261	Thermodynamics of Life Processes	6	
LB1270	Biotechnology and Society	4	
LB1280	Biotechnology, practical basics	5	
LB1300	Calculus 1 (incl. aansluitmodule)	4	
LB1305	Calculus 2	3	
LB1310	Calculus 3	3	
LB1512TU	Biotechnology, Theory	6	
LB2512	Biochemie 2	3	
2e jaar BSc LST 2022	2nd year BSc LST 2022		
LB1211	Statistics	3	
LB2031	Molecular Cell Biology & Immunology	8	
LB2041	Molecular Genetics & Gene Technology	9	
LB2055	Molecular Physics	4	
LB2261	Statistical Thermodynamics	4	
LB2301	Organische Chemie 2	6	
LB2311	Chemische Biologie	3	
LB2532	Transport Phenomena in the Life Sciences	6	
LB2630	Molecular Analytical Methods	4	
LB2763	Microbial Physiology	5	
LB2771	Metabolic Engineering	4	
LB2791-P	Practicals Microbial biotechnology	4	
3e jaar BSc LST 2022			
LB2291	Bio-informatics	8	
LB2505-P	Bachelor Research Project	17	
LB2611	Design of Sustainable Biotechnological Processes	5	
Minor			

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Bachelor LST 2022

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1e jaar BSc LST 2022

LB1045	Organic Chemistry 1	6
Responsible Instructor	Dr. S. van der Vorm	
Instructor	Dr. S. van der Vorm	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

LB1053	Cell biology & Biochemistry: an introduction	5
Responsible Instructor	M. Ubbink	
Instructor	M. Ubbink	
Instructor	Dr. M.C. van Eijk	
Instructor	R. Boot	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Literature and Study Materials	Verplichte boeken: Molecular Biology of The Cell, 6th edition van Alberts et al. Lehninger Principles of Biochemistry, 7th edition van Nelson en Cox De presentaties van de colleges van het Celbiologie worden gepubliceerd op de de Blackboard course. Voor het deel Biochemie 1 wordt een syllabus uitgereikt en uitwerkingen van de vragen verschijnen op de Blackboard course.	

LB1231	Life Sciences	8
Responsible Instructor	Dr. M.C. van Eijk	
Instructor	Drs. A.E. Kam	
Instructor	Dr. W.R.M. Schlaman	
Instructor	Dr. M.C. van Eijk	
Instructor	Dr. M.H. de Smit	
Instructor	L.J.C. Jeuken	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	

LB1251	Quantitative Molecular Biology	4
Responsible Instructor	R.T. Dame	
Instructor	R.T. Dame	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	

LB1260	Introduction to Algorithmics and Programming using Python	3
Responsible Instructor	R. Gonzalez Cabaleiro	
Instructor	R. Gonzalez Cabaleiro	
Instructor	Dr.ir. C. Haringa	
Contact Hours / Week x/x/x/x	0/0/16/16	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Course Contents	The course provides an introduction to designing algorithms, performing calculations and coding computer programs in Python - computer language.	
Study Goals	The course consists of 4 lectures (2 h each) and 10 computer-assisted practical sessions (2 h each). Homework exercises and assignments will be delivered. The course aims at developing the mindset for solving numerical problems and constructing logical algorithms. Applications and code will be written in Python, currently a very widespread computer language. After the course the student should be able to: - understand and use basic programming concepts (conditional statements, loops, variables, arrays, indexing, input/output, functions) - develop simple numerical algorithms - use basic functions of Python and write own functions - display results in a graphical form (plotting) - perform code debugging	
Education Method	- 8 h lectures on programming techniques (4 sessions x 2 h) - 20 h computer exercise (10 sessions x 2 h) Group programming project.	
Computer Use	Practical sessions in computer rooms, one student per computer. Python programming language.	
Assessment	During the course: One short quizz containing knowledge based questions, pass/fail, 2.5% of final One short quizz on a programming probleme (skill based), pass/fail, 2.5% of final Finished lab exercises (2 submissions), pass/fail, 5 % of final Group programming project (groups of 3), 20 % of final End of the course: Computer-based exam (Python programming tasks), in exam period, 70% of final grade. On site o proctored.	
Exam Hours	3 hours exam with computer on developing numerical algorithms and programming in Python.	
Special Information	Email address: Rebeca.Gonzalez-Cabaleiro@glasgow.ac.uk (until february 2021)	
Tags	Programming Programming concepts	
Studyload/Week	About 8 h per week, during 12 weeks	
Location	Delft	

LB1261	Thermodynamics of Life Processes	6
Responsible Instructor	Dr.ir. R. Kleerebezem	
Instructor	Dr.ir. R. Kleerebezem	
Instructor	Dr.ir. C. Haringa	
Education Period	4	
Start Education	4	
Exam Period	4 4A 4B	
Course Language	Dutch	

LB1270	Biotechnology and Society	4
Responsible Instructor	Drs. L. Asveld	
Instructor	Drs. L. Asveld	
Instructor	I.M. Schouten	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Course Contents	- Maatschappelijke context en implicaties van biotechnologische ontwikkelingen (waaronder biofarmaceutische technologie) - Academische vaardigheden (mondeling presenteren)	

LB1280	Biotechnology, practical basics	5
Responsible Instructor	Prof.dr.ir. P.A.S. Daran-Lapujade	
Instructor	Prof.dr.ir. P.A.S. Daran-Lapujade	
Instructor	Ing. A.F. de Hulster	
Instructor	S.A. Pilon	
Contact Hours / Week x/x/x/x	0/0/136	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch English	
Course Contents	This practicum aims to teach you some of the fundamentals of biotechnology so that, whatever your eventual specialty, you will be able to function in a multi-disciplinary team and understand what your research partners are talking about. The practicum is divided into microbiology and fermentation to teach how to work with microbes following biosafety rules, how to analyse and interpret data and write scientific reports.	
Study Goals	Microbiology After this practicum you should be able to: LO 1: Work in a microbiological laboratory according to the Safe Microbiology Techniques and Regulations. More precisely this means that you will be able to: LO 1.1: describe safety classes of microorganisms LO 1.2: describe the corresponding safety demands, regulations and procedures, and work according to them LO 1.3: work according to Good Laboratory Practices LO 2: Analyse, explain and evaluate results LO 3: Design basic experiments LO 4: Work in teams Fermentation LO 5: Know how a fermenter works and how to determine the oxygen transfer capacity of a fermenter LO 6: Analyse, explain and evaluate fermentation results LO 7: Use measured concentrations and the stoichiometry to make balances LO 8: Check the accuracy of measurements by means of balances	
Education Method	During the practicum students will work in duos and perform experiments in microbiology and fermentation. During 18 days spread over 9 weeks students will work in the lab, analyse and interpret data, and write a report about their research. Individual reports will be written. The practicum takes place at TU Delft in building 58.	
Assessment	Student's performance will be assessed individually for both the microbiology and fermentation modules at the level of 50% per module. For each module, the grade is composed of a report and the evaluation of lab skills: - microbiology: 50 % report, 50% lab skills - fermentation: 75% report, 25% lab skills	

LB1300	Calculus 1 (incl. aansluitmodule)	4
Responsible Instructor	M. Roelands	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

LB1305	Calculus 2	3
Responsible Instructor	Dr. E.M. Blokhuis	
Instructor	Dr. E.M. Blokhuis	
Instructor	Dr. G.J.A. Sevink	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	
Course Contents	Differential Equations; Infinite Series	
Study Goals	Understanding concepts and being able to solve certain relevant differential equations. Understanding concepts and being able to determine whether an infinite series converges. Understanding and being able to construct Taylor series.	
Education Method	hoorcolleges; werkcolleges; begeleide zelfstudie	
Assessment	written exam	

LB1310	Calculus 3	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Instructor	Dr.ir. T.M.L. Janssen	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	Dutch English	

LB1512TU	Biotechnology, Theory	6
Responsible Instructor	Prof.dr. J.T. Pronk	
Instructor	Prof.dr. J.T. Pronk	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Required for	Microbiële Fysiologie LB2763, Geïntegreerd Practicum Biotechnologie LB2791-p Metabolic Engineering LB2771 Ontwerpen van Duurzame Biotechnologische Processen LB2611	
Expected prior knowledge	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Course Contents	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Study Goals	This course is taught in Dutch, please switch to Dutch language mode to read course informationin Dutch	
Education Method	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Literature and Study Materials	This course is taught in Dutch, please switch to Dutch language mode to read course information exercises (also in Dutch)	
Reader	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Assessment	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Remarks	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Continuing Courses	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Location	This course is taught in Dutch, please switch to Dutch language mode to read course information	

LB2512	Biochemie 2	3
Responsible Instructor	Dr.ir. P.L. Hagedoorn	
Instructor	Dr.ir. P.L. Hagedoorn	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	

Year	2022/2023
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2e jaar BSc LST 2022

LB1211	Statistics	3
Responsible Instructor	V. Masarotto	
Instructor	V. Masarotto	
Contact Hours / Week x/x/x/x	29/0/0/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Having passed a calculus course is not necessary, but it's strongly recommended.	
Course Contents	The course introduces key concepts of probability and applies them to the formulation of key notions of statistics. Random variables and their distributions (with examples and application of both discrete and continuous variables), expectation and variance. The law of large numbers and the central limit theorem. Statistical testing to check whether two statistical objects are the same, with as special examples linear regression and the t-test.	
Study Goals	Insight in modelling uncertainty and measurement error. Rules for probabilities Expectation and variance Limit theorems for sums of random variables Statistical testing Confidence intervals	
Education Method	Lectures and problem sessions. **Being exposed to examples and solved exercises plays a key role in understanding the material, which is why the course contains solved exercises. However, in order to give proper time to the lessons, exercises will be solved and showed at the blackboard ONLY during the werkcolleges. Attendance is therefore very strongly encouraged**	
Computer Use	Some exercises using a computer package may be incorporated.	
Literature and Study Materials	The course is based on self-contained slides and recording from the lectures, together with some worked out exercises. A syllabus (ongeveer 60 paginas) in Dutch from prof. Aad Van der Vaart from previous years will be made available on brightspace. Het volgende boek kan worden gebruikt voor verder lezen: F.M. Dekking, C. Kraaikamp, H.P. Lopuhaa and L.E. Meester, A Modern Introduction to Probability and Statistics. Springer (2006).	
Assessment	Written exam. First date: 24.10.2022, 13-16. Retake: 9.11.2022, 13-16.	
Special Information	The course introduces novel concepts of Probability and Statistics. Every lesson builds on the previous one. Werkcolleges are an essential part for the completion and understanding of the course. Examples and exercises will be showed, discussed, solved in details ONLY during the werkcollege. Teacher Information: Dr. Valentina Masarotto Assistant Professor Mathematical institute leiden university p.o. box 9512 2300 RA Leiden office 221, Niels Bohrweg 1	
Location	Leiden	

LB2031	Molecular Cell Biology & Immunology	8
Responsible Instructor	Dr. M.C. van Eijk	
Instructor	Dr. M.C. van Eijk	
Instructor	Dr. S.I. van Kasteren	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

LB2041	Molecular Genetics & Gene Technology	9
Responsible Instructor	Dr. M.C. van Eijk	
Instructor	Dr. M.C. van Eijk	
Instructor	Dr. M.H. de Smit	
Education Period	4	
Start Education	4	
Exam Period	4 4A 4B	
Course Language	Dutch	

LB2055	Molecular Physics	4
Responsible Instructor	M. Huber	
Instructor	M. Huber	
Education Period	3 4	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	Voorkennis: Organische Chemie, Quantitatieve Moleculaire Biologie, Calculus	
Literature and Study Materials	Physical Chemistry for the Life Sciences 3rd edition, Peter Atkins, Juilo de Paula, Publisher: Oxford University Press ISBN-10 3rd edition is: 9780198830108.	
Special Information	Docent emailadres: huber@physics.leidenuniv.nl	
Studyload/Week	A brief calculation of the course load, broken down by: - Total course load for the course (number of EC x 28 hours), for a course of 5 EC is 140 hours, for 10 EC 280. - Hours spent on attending lectures and seminars (eg 2 hours per week x 14 weeks = 28 hours) - Time for studying the compulsory literature (as a possible criterion approx. 7 pages per hour with deviations up and down depending on the material to be studied) (if applicable) time for completing assignments, whether in preparation at the college - (If applicable) time to write a paper (including reading / research) Totale hoeveelheid uren 4 EC	

LB2261	Statistical Thermodynamics	4
Responsible Instructor	Dr. E.M. Blokhuis	
Instructor	Dr. E.M. Blokhuis	
Contact Hours / Week x/x/x/x	0/0/53/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	

LB2301	Organische Chemie 2	6
Responsible Instructor	Dr. S. van der Vorm	
Instructor	M. Overhand	
Instructor	Dr. S. van der Vorm	
Contact Hours / Week x/x/x/x	0/47/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	

LB2311	Chemische Biologie	3
Responsible Instructor	Prof. M. van der Stelt	
Instructor	Dr. R.J.B.H.N. van den Berg	
Instructor	Dr. S.I. van Kasteren	
Instructor	Prof. M. van der Stelt	
Instructor	Dr. A.P.A. Janssen	
Contact Hours / Week x/x/x/x	0/0/29/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	

LB2532	Transport Phenomena in the Life Sciences	6
Responsible Instructor	Dr. M.E. Klijn	
Instructor	Prof.dr.ir. M. Ottens	
Instructor	Dr. M.E. Klijn	
Contact Hours / Week x/x/x/x	0/0/0/96	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Literature and Study Materials	Janssen L B P M and M M C G Warmoeskerken, Transport Phenomena Data Companion.	
Location	Delft	

LB2630	Molecular Analytical Methods	4
Responsible Instructor	Dr. K. Djanashvili	
Instructor	M.J.F. Strampaad	
Instructor	Dr.ir. C. Haringa	
Instructor	Dr. M. Pabst	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

LB2763	Microbial Physiology	5
Responsible Instructor	R.J. van Tatenhove-Pel	
Instructor	R. Mans	
Instructor	R.J. van Tatenhove-Pel	
Contact Hours / Week x/x/x/x	0/43/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	
Expected prior knowledge	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Course Contents	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Study Goals	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Education Method	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Literature and Study Materials	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Assessment	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Studyload/Week	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Continuing Courses	This course is taught in Dutch, please switch to Dutch language mode to read course information	
Location	This course is taught in Dutch, please switch to Dutch language mode to read course information	

LB2771	Metabolic Engineering	4
Responsible Instructor	R.J. van Tatenhove-Pel	
Instructor	Dr. W.M. van Gulik	
Contact Hours / Week x/x/x/x	0/0/33/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	Biotechnology theory, Microbial Physiology	
Course Contents	This course focuses on quantitative approaches to assess the performance of industrial microorganisms, to identify targets for their improvement through metabolic engineering and to calculate the impact of these metabolic engineering efforts on the economics of the fermentation process. Subjects which will be addressed are: - Black box modeling of growth and product formation in microorganisms - Construction of ODE based models in python for simulation of batch, fed-batch and chemostat cultivations - Defining the overall reaction for a fermentation process (process reaction) - Construction and application of metabolic structured models - Stoichiometric modeling and flux balance analysis - Metabolic engineering strategies for lowering costs of fermentation processes - Case study on the design of a fermentation process using python	
Study Goals	Learn to apply quantitative approaches to identify targets for improvement of fermentation processes through metabolic engineering By the end of this course, you should be able to: LO1: Explain the following quantitative models of microorganisms: black-box, metabolically structured and stoichiometric models. LO2: Apply dynamic simulations of fermentation processes (batch, fed-batch, chemostat, retentostat) LO3: Analyze the performance of industrial microorganisms using black-box models and metabolic models, by hand and by using Python. LO4: Design metabolic engineering targets for lowering costs in fermentation processes, based on a short process description. LO5: Describe how quantitative models of microorganisms can be used within the life sciences and adjacent disciplines.	
Education Method	Lectures & exercises	
Literature and Study Materials	Lecture slides, exercises plus detailed answers	
Assessment	Case study/group assignment(s) (25% of final grade) and closed book exam (75% of final grade). The grade for the case study/group assignment(s) will only count for the final grade if the grade for the exam is minimally 5.0.	
Continuing Courses	Analysis of Metabolic Networks, Fermentation Technology and Environmental Biotechnology	
Location	Delft	

LB2791-P	Practicals Microbial biotechnology	4
Responsible Instructor	Dr. D.G.G. McMillan	
Instructor	S.A. Pilon	
Contact Hours / Week x/x/x/x	0/10 dgn/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	Dutch English	
Expected prior knowledge	Biotechnologische Basistechnieken (LB1280) and Biotechnologie (LB1512TU).	
Course Contents	This practicum is a laboratory scale process set on a background of what is typically found in the biotechnology industry. The exercise utilises the microbial production of an extracellular protein (inulinase) in fed batch fermentation, downstream processing and formulating steps (microfiltration, ultrafiltration and immobilisation). Lastly, the protein will be used as an immobilised biocatalyst in a packed bed reactor for the production of the natural sweetener fructose.	
Study Goals	1. To provide an in-depth understanding of the different ways of carrying out fermentation processes (batch, fed-batch, chemostat) with their advantages and disadvantages. 2. To obtain practical experience in designing and carrying out a fed-batch process. 3. To develop an appreciation of enzyme production in microorganisms for down stream application. 4. To develop an appreciation of the interference that contaminants can have on both DSP and enzyme utilization. 5. To develop an understanding of the mechanisms by which commonly used separation techniques operate. 6. To obtain practical experience in purifying a fermentation broth to obtain an enzyme solution suitable for utilization. 7. To examine how a flow model for a packed bed reactor can be determined experimentally. 8. To assess whether the models from day 1 can produce a good description of the behaviour of a continuous reactor and to allow operation of the reactor under optimal conditions.	
Education Method	Practicum.	
Computer Use	Data analysis: MS Excel, Python Text processing: MS Word (optionally: LaTeX) Reference manager: free choice, e.g. Mendeley, OneNote, EndNote	
Literature and Study Materials	Practical Manual.	
Assessment	Students are assessed on: 1) Demonstrating how to apply taught knowledge in a practical context. This is achieved through examination of laboratory skills (i.e. measurement accuracy and dedication during the practicum). This element makes up 25% of the final mark; 2) Quality of experimental evaluation; report quality in terms of legibility, writing style, and interpretation of results. This element makes up 75% of the final mark; The different parts of the practicum are weighed as follows: fermentation 40%, downstream processing 30%, bioconversion 30%. If the assistants are not Dutch-speaking the report has to be written in English.	
Enrolment / Application	Subscribe at least three weeks before the start of the practicum via the practicum subscribe system on the Brightspace course of LST.	
Continuing Courses	This practicum serves as a preparation for Metabolic Engineering, Industrial Biotechnology and the Bachelor End Project.	
Location	Applied Sciences (Building 58), Delft	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Life Science and Technology

3e jaar BSc LST 2022

LB2291	Bio-informatics	8
Responsible Instructor	Prof.dr.ir. M.J.T. Reinders	
Instructor	Prof.dr.ir. M.J.T. Reinders	
Instructor	S.R.J. Geibel	
Contact Hours / Week x/x/x/x	0/0/65/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch English	
Expected prior knowledge	Basic understanding of biochemistry and python	
Course Contents	Bioinformatics course consists of two parts: structure analysis and data analysis. Structure analysis: A new generation of structure prediction programs has shown unprecedented accuracy and is heralding a new era in structural biology. This course will introduce the principles of structure prediction programs and contrast them to experimental structure determination methods such as single-particle cryo-electron microscopy (cryo-EM), X-ray crystallography, and nuclear magnetic resonance spectroscopy (NMR). We will discuss the current limitations of these methods and show how to evaluate the quality of experimental and predicted structural models. Students will learn how to analyze and explore the function of unknown protein structures. Building on this, we will introduce rational drug design as an application benefiting from accurate structures and knowledge of their active sites. Data analysis: statistical testing, clustering, classification	
Study Goals	The students following this course will have: - basic knowledge bioinformatics methodologies. and they will be trained to: - apply basic bioinformatics tools on real data - explain the theoretical principles behind experimental structure determination and structure prediction methods - explain the concepts of artificial intelligence driven learning models - evaluate the quality of predicted and experimental protein structures - analyze protein structures and how to explore their functions - discuss rational drug design and virtual drug screening approaches	
Education Method	Structure analysis part: Lectures and practical exercises (computer practicals) Data analysis part: Lectures, assignments (computer practicals), project	
Literature and Study Materials	Reading material is provided during the lectures and at Brightspace. You will use your own laptop, computers in computer rooms, specialized and open source software.	
Assessment	A final written exam, exercises and assignments. -Structure analysis: - written exam with short and essay questions (grading). The grade for structure analysis part is equal to the grade for the (structure analysis part of the) final exam. - Data analysis: assignments (passing), projects (grading), exams (grading) The grade for the data analysis part is composed of: 1) submission of answers to practicals 2) working version of the programming assignment 3) execution of the Quizzes 4) 20% : grade for poster first two modules (statistics) 5) 20% : grade for poster last two modules (clustering, classification) 6) 60% : grade for (data analysis part of) the final exam The final grade of the course is the average of the grade for the sequence structure analysis part and the grade for data analysis part. The final exam, covering the course content of both parts, is at 6 April 2023 (Delft). The retake, covering the entire course content (both parts), counts 100% is at 21 April 2023 (Delft).	
Remarks	Contacts for structure analysis part: Sebastian Geibel: s.r.j.geibel@lic.leidenuniv.nl and Steffen Brünle: s.bruehle@lic.leidenuniv.nl Address: Gorlaeus Building, Einsteinweg 55, 2333 CC Leiden, Room: DM.2.14	

LB2505-P	Bachelor Research Project	17
Course Coordinator	Dr. W.R.M. Schlaman	
Contact Hours / Week x/x/x/x	2 weken (5 dgn/wk)	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	Dutch English	
Expected prior knowledge	All courses from the major bachelor Life Science and Technology (see entry requirements).	
Books	'On Being a Scientist: A guide to responsible conduct in Research.' 3th edition. FREE DOWNLOAD: https://www.nap.edu/catalog/12192/on-being-a-scientist-a-guide-to-responsible-conduct-in	
Prerequisites	Students enrolled in the bachelor programme LST from September 2018 onwards: - All components of the first year should be completed (60 ec), - At least 45 ec of components of the 2d year curriculum, including all practical courses. Students enrolled in the bachelor programme LST in September 2017: - All components of the first year should be completed (propedeuse exam, 60 ec), - At least 30 ec of components of the 2d year curriculum, including all practical courses. Students enrolled in the bachelor programme LST from September 2013 until September 2016: - All components of the first year should be completed (propedeuse exam, 60 ec), - At least 30 ec of components of the 2d year curriculum, including all practical courses and the design project (ODBP). Students enrolled in the bachelor programme LST since September 2007 until September 2012: - All components of the first year should be completed (propedeuse exam, 60 ec), - At least 30 ec of components of the 2e year curriculum, including all practical courses and the design project (MVO) and the course 'Mondeling Presenteren'. Students enrolled in the bachelor programme LST in September 2006 and before: - All components of the first year should be completed (propedeuse exam, 60 ec), - All practical components of the 2d and 3d year curriculum should be completed, - The 3d year components 'Metabolic Engineering', 'Industrial Biotechnology', 'Statistische Thermodynamica' and 'Bio-informatica 1' should be completed. All students: - Approval of the academic counselor to start the project (as shown on the application form). - Submission of the filled-out application form ('Aanmeldformulier'), undersigned by the student, responsible supervisor(s) and the academic counselor prior to the start of the project.	
Continuing Courses	Master programme LST or other connecting master programmes	
Location	Leiden or Delft	

LB2611	Design of Sustainable Biotechnological Processes	5
Responsible Instructor	Dr. J.A. Posada Duque	
Instructor	Dr. J.A. Posada Duque	
Instructor	Dr. L. Jourdin	
Contact Hours / Week x/x/x/x	0/0/90/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	The first three semesters of the BSc Life Science & Technology	
Course Contents	Integration of various scientific and technological approaches, and their application to the design of sustainable biotechnological processes at industrial scale. Lectures from teachers and distinguished guests will focus on different microorganisms, products and processes. In a group project, you will apply the learned knowledge to redesign a biotechnological process aiming to improve its economics, safety and sustainability aspects. Results of the redesign project and learnings from guest lectures will be disseminated (and graded) through a report, an executive summary and two presentations.	
Study Goals	Upon completion of this course the student will be able to: - Understand and evaluate the use of microorganisms (and enzymes) for industrial production, at the level of the microbial physiology, product formation and production system; - Describe an existing industrial biotechnological production processes by means of block diagrams and mass balances; - Analyse a biotechnological production method based on its economic metrics, sustainability principles and safety basics.	
Education Method	Instructive course lectures by the teachers, guest lectures, and group work sessions for the design project.	
Literature and Study Materials	Literature and study materials will be made available or indicated during the course.	
Assessment	The overall grade is composed of three aspects: - Report of the standard process of the case study (35%); - Executive summary and presentation of the redesigned/improved process (45%); - Presentation of reflection on guest lectures (20%).	
Location	TUDelft	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Life Science and Technology

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Department	Financial control
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